Name ____ Date ___________ Period _____

Parent Functions III: 1/x and Asymptotes

Algebra II | Unit 1 | Day 6 | TEK A2.2.A

Objective: I can graph the reciprocal parent function, identify its vertical and horizontal asymptotes, and describe its domain, range, and symmetry.

Vocabulary

Asymptote — a line the graph gets ___________________ to but never settles on.

Vertical asymptote — a vertical line the graph _________________. It sits where the function is undefined.

Horizontal asymptote — a horizontal line describing end behavior. It ________ be crossed in other functions.

Reciprocal — the reciprocal of a is ________. Zero has ______ reciprocal.

Part 1 — Build the Table for f(x) = 1/x

Fill in the outputs. One input has no output at all.

x	-2	-1	-1/2	0	1/2	1	2
f(x) = 1/x							

Why is x = 0 undefined? ____

Part 2 — What Happens Near Zero and Far Out

Follow the pattern in each list, then describe it.

x	1	0.5	0.1	0.01
1/x	1	2	10	100

As x gets close to 0, the outputs grow ___________________. So there is a vertical asymptote at __________.

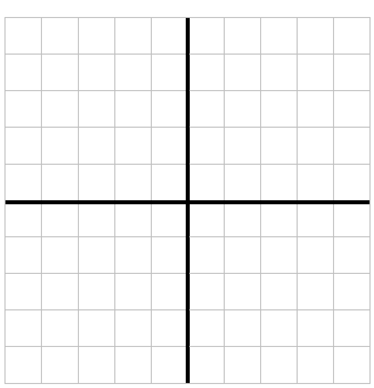
x	1	10	100	1000
1/x	1	0.1	0.01	0.001

As x gets very large, the outputs approach ________. So there is a horizontal asymptote at __________.

Part 3 — Graph It

Each square is 1 unit. Plot your table points. You will get two separate branches - do not connect them across the y-axis.

f(x) = 1/x



Record what you see.

Vertical asymptote: ______________

Horizontal asymptote: ______________

How many branches? ________

Which quadrants? ______________

Any intercepts? ______________

Draw both asymptotes as dashed lines.

Part 4 — Key Attributes of f(x) = 1/x

Domain: ___________________________ set notation ___________________________

Range: ___________________________ interval ___________________________

Symmetry: reflectional across ______________________, and rotational ________ about the origin.

Maximum or minimum? __________________ Intercepts? __________________

This is the first parent function with __________________ and the first whose domain has a __________ in it.

Part 5 — True or False

1. A graph may cross a vertical asymptote. ________________
2. A graph may cross a horizontal asymptote. ________________
3. f(x) = 1/x has a y-intercept. ________________
4. The domain of 1/x is every real number. ________________

Practice

Evaluate and explain. Show your work where there is room.

1. Evaluate: 1/4 = ____________ 1/(-2) = ____________ 1/(1/5) = ____________

2. Solve 1/x = 4. x = ________________ Solve 1/x = -1. x = ________________

3. Is there any x with 1/x = 0? Explain.

____

____

4. As x grows very large and negative, what happens to 1/x? _______________________________

5. Fill in the pattern.

x	-0.1	-0.01	0.01	0.1
1/x				

Approaching 0 from the left the outputs go toward ____________, from the right toward ____________.

6. Why is 1/x the only parent function so far with a gap in its domain?

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Exit Ticket

A function has no intercepts, a vertical asymptote at x = 0, a horizontal asymptote at y = 0, and two branches in Quadrants I and III.

Name it: _________________________ Domain in set notation: ___________________________